The four stage low-pass filter is comparable to the Blackman and Gaussian filters (relatives of the moving average, Chapter 15), but with a much faster execution speed. The design equations for a four stage low-pass filter are:

EQUATION 19-6
Four stage low-pass filter. These equations provide the "a" and "b" coefficients for a cascade of four single pole low-pass filters. The relationship between x and the cutoff frequency of this filter is given by Eq. 19-5, with the 2π replaced by 14.445.

$$\begin{aligned} a_0 &= (1-x)^4 \\ b_1 &= 4x \\ b_2 &= -6x^2 \\ b_3 &= 4x^3 \\ b_4 &= -x^4 \end{aligned}$$

Narrow-band Filters

A common need in electronics and DSP is to isolate a narrow band of frequencies from a wider bandwidth signal. For example, you may want to eliminate 60 hertz interference in an instrumentation system, or isolate the signaling tones in a telephone network. Two types of frequency responses are available: the *band-pass* and the *band-reject* (also called a **notch filter**). Figure 19-6 shows the frequency response of these filters, with the recursion coefficients provided by the following equations:

EQUATION 19-7
Band-pass filter. An example frequency response is shown in Fig. 19-6a. To use these equations, first select the center frequency, f, and the bandwidth, BW. Both of these are expressed as a fraction of the sampling rate, and therefore in the range of 0 to 0.5. Next, calculate R, and then K, and then the recursion coefficients.

$$\begin{aligned} a_0 &= 1-K \\ a_1 &= 2(K-R)\cos(2\pi f) \\ a_2 &= R^2-K \\ b_1 &= 2R\cos(2\pi f) \\ b_2 &= -R^2 \end{aligned}$$

EQUATION 19-8
Band-reject filter. This filter is commonly called a notch filter. Example frequency responses are shown in Fig. 19-6b.

$$\begin{aligned} a_0 &= K \\ a_1 &= -2K\cos(2\pi f) \\ a_2 &= K \\ b_1 &= 2R\cos(2\pi f) \\ b_2 &= -R^2 \end{aligned}$$

where:

$$K = \frac{1 - 2R\cos(2\pi f) + R^2}{2 - 2\cos(2\pi f)}$$

$$R = 1 - 3BW$$